WoSM Academy

M201-F27 · Calculus II

Taylor series

Taylor and Maclaurin series; remainder estimates

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WoSM Academy

Lecture 06 · Unit 6 · Fall 2027

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Last time

- A power series $\sum c_k (x-a)^k$ converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$.
- Inside, we may differentiate and integrate it term by term.
- If $f(x) = \sum c_k (x-a)^k$ near a, then $c_k = \frac{f^{(k)}(a)}{k!}$.
- From the geometric series: $\ln(1+x)$ and $\arctan x$ for $|x| < 1$. From $E' = E$: $e^x = \sum \frac{x^k}{k!}$.

Today we start from a function f and ask: is it a power series, and which one?

This lecture

01 Taylor series

02 Taylor's theorem

03 The classical series

04 Working with Taylor series

05 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Sections 6.3 and 6.4.

01

Taylor series

The only candidate

If f is a power series around a, its coefficients must be $\frac{f^{(k)}(a)}{k!}$ (Lecture 5).

Definition

Let f have derivatives of every order at a. The **Taylor series** of f at a is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

At $a = 0$ it is also called the **Maclaurin series**.

Its partial sums are the **Taylor polynomials** of Calculus I (Unit 17), $T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$.

Two questions

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2. If it converges at x, is its sum $f(x)$?

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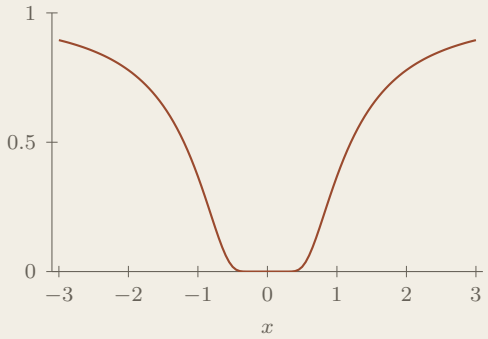
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Definition

f is **analytic** at a if its Taylor series at a converges to f on some interval $(a-r, a+r)$, $r > 0$.

Polynomials, e^x, sin, cos, ln, rational functions and $(1+x)^\alpha$ are analytic wherever they are defined. But not every smooth function is.

A smooth function that is not analytic



$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- For $x \neq 0$ every derivative has the form $p(\frac{1}{x})\, e^{-1/x^2}$ with p a polynomial; it tends to 0 as $x \to 0$. One shows $f^{(k)}(0) = 0$ for all k.
- So the Taylor series at 0 is $0 + 0x + 0x^2 + \cdots$. It converges everywhere — to 0, not to $f(x)$.

The graph is so flat at 0 that no derivative can see it rise.

02

Taylor's theorem

Taylor's theorem

Theorem (Taylor's theorem, Lagrange remainder — Calculus I)

If f is $n+1$ times differentiable on an interval containing a and x, then

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1} \qquad \text{for some } \xi \text{ between } a \text{ and } x.$$

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Corollary (Remainder estimate)

If $|f^{(n+1)}(t)| \le M$ for all t between a and x, then $|R_n(x)| \le \frac{M\,|x-a|^{n+1}}{(n+1)!}$.

We do not know ξ; a bound for $f^{(n+1)}$ on the interval is enough.

The remainder as an integral

Theorem

If $f^{(n+1)}$ is continuous between a and x, then $R_n(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!}(x-t)^n \, \mathrm{d}t$.

Proof.

Induction on n ($n = 0$: the Fundamental Theorem of Calculus). By parts,

$$\int_a^x \frac{f^{(n)}(t)}{(n-1)!}(x-t)^{n-1} \, \mathrm{d}t = \frac{f^{(n)}(a)}{n!}(x-a)^n + \int_a^x \frac{f^{(n+1)}(t)}{n!}(x-t)^n \, \mathrm{d}t,$$

that is, $R_{n-1}(x) = \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$. □

A useful limit

Lemma

For every $A \geq 0$: $\dfrac{A^n}{n!} \to 0$.

Proof: these are the terms of $\sum \frac{A^n}{n!}$, which converges (ratio test).

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Corollary

If $|f^{(n)}(t)| \leq M$ for all n and all t in an interval I containing a, then f equals its Taylor series on I.

Proof: $|R_n(x)| \leq M\frac{|x-a|^{n+1}}{(n+1)!} \to 0$: the factorial beats any fixed power.

03

The classical series

The exponential function

All derivatives of e^x are e^x. Between 0 and x, $e^{\xi} \le e^{|x|}$, so

$$|R_n(x)| \le e^{|x|} \frac{|x|^{n+1}}{(n+1)!} \to 0.$$

Again: $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for every x — a second proof.

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Example (Computing e)

$e = \sum_{k=0}^{\infty} \frac{1}{k!}$, and $|R_n(1)| \le \frac{e}{(n+1)!} < \frac{3}{(n+1)!}$. With $n = 10$: $\sum_{k=0}^{10} \frac{1}{k!} = 2.7182818\ldots$, with an error below $\frac{3}{11!} \approx 7.5 \cdot 10^{-8}$.

e is irrational

Theorem

e is not a rational number.

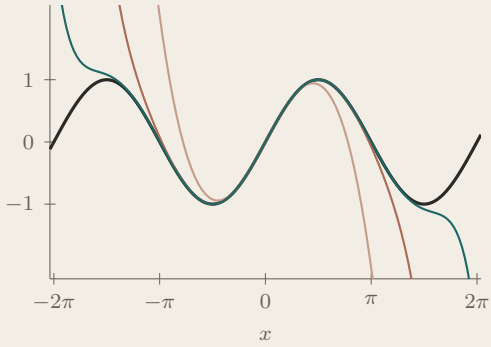
Proof.

Suppose $e = \frac{p}{q}$ with integers $p, q \geq 1$. Then $N = q!(e - \sum_{k=0}^{q} \frac{1}{k!}) = p\,(q-1)! - \sum_{k=0}^{q} \frac{q!}{k!}$ is an integer, and $N > 0$. But

$$N = \sum_{k>q} \frac{q!}{k!} = \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \cdots < \frac{1}{q+1} + \frac{1}{(q+1)^2} + \cdots = \frac{1}{q} \leq 1.$$

An integer strictly between 0 and 1 does not exist. □

Sine and cosine



The derivatives cycle through $\sin, \cos, -\sin, -\cos$, so all are bounded by 1, and $|R_n(x)| \le \frac{|x|^{n+1}}{(n+1)!} \to 0$. For every x:

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

Each new term keeps the polynomial close to $\sin x$ on a wider interval: T_3, T_7, T_{11}.

How a computer might compute $\sin x$

Example

$\sin 0.1 \approx 0.1 - \frac{0.1^3}{6} = 0.0998333\ldots$ Since $T_3 = T_4$ for sin, the error is at most $\frac{0.1^5}{5!} \approx 8.4 \cdot 10^{-8}$.

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- Mathematical libraries typically first reduce x to a small interval, such as $|x| \le \frac{\pi}{4}$, using $\sin(x + 2\pi) = \sin x$ and similar identities.
- On $|x| \le \frac{\pi}{4}$ the Taylor polynomial of degree 15 has error at most $\frac{(\pi/4)^{17}}{17!} < 5 \cdot 10^{-17}$ — below the precision of a double.
- Real libraries use polynomials tuned to the whole interval rather than Taylor polynomials, but the principle is the same: a polynomial, plus a proof of how close it is.

$\ln 2$, at last

For $f(x) = \ln(1+x)$: $\quad f^{(n+1)}(t) = \dfrac{(-1)^n\, n!}{(1+t)^{n+1}}.$

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$$|R_n(x)| = \frac{n!}{(1+\xi)^{n+1}} \cdot \frac{x^{n+1}}{(n+1)!} \le \frac{1}{n+1} \to 0.$$

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Theorem

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

The value promised in Lecture 3. For $-1 < x < 0$ this bound is too weak, since $1 + \xi < 1$; there Lecture 5 already gave us the series.

$\frac{\pi}{4}$, at last

Stop the geometric series after $n+1$ terms and keep the rest exactly:

$$\frac{1}{1+t^2} = \sum_{k=0}^{n} (-1)^k t^{2k} + \frac{(-1)^{n+1} t^{2n+2}}{1+t^2}.$$

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Integrate from 0 to 1:

$$\frac{\pi}{4} = \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} + E_n, \qquad |E_n| \le \int_0^1 t^{2n+2}\,\mathrm{d}t = \frac{1}{2n+3} \to 0.$$

Letting $n \to \infty$: $\dfrac{\pi}{4} = 1 - \dfrac{1}{3} + \dfrac{1}{5} - \dfrac{1}{7} + \cdots$ (Leibniz). It converges slowly (Lecture 3). Machin's formula $\frac{\pi}{4} = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$ converges fast: with it, Machin computed 100 decimal places of π in 1706.

The binomial series

For $\alpha \in \mathbb{R}$ let $\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}$ and $\binom{\alpha}{0} = 1$.

Theorem (Binomial series)

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k \quad \text{for } |x| < 1.$$

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Proof.

For $\alpha \in \{0, 1, 2, \dots\}$ the series stops: the binomial theorem. Otherwise $\left|\binom{\alpha}{k+1}/\binom{\alpha}{k}\right| = \frac{|\alpha-k|}{k+1} \to 1$, so $R = 1$. Let g be the sum. As $(k+1)\binom{\alpha}{k+1} = (\alpha-k)\binom{\alpha}{k}$, term by term $(1+x)\,g' = \alpha\, g$. So $\big(g(1+x)^{-\alpha}\big)' = 0$, and $g(0) = 1$. □

Binomial series: examples

- $\alpha = \frac{1}{2}$: $\quad \sqrt{1+x} = 1 + \dfrac{x}{2} - \dfrac{x^2}{8} + \dfrac{x^3}{16} - \cdots$
- $\alpha = -1$: $\quad \dfrac{1}{1+x} = 1 - x + x^2 - \cdots$ $\quad$ (the geometric series again)
- $\alpha = -\frac{1}{2},\ x \to -4x$:
  $\dfrac{1}{\sqrt{1-4x}} = \sum_{k=0}^{\infty} \binom{2k}{k} x^k = 1 + 2x + 6x^2 + 20x^3 + 70x^4 + \cdots \qquad (|x| < \frac{1}{4})$

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Remark

$\binom{2k}{k}$ counts the paths from $(0,0)$ to (k,k) with unit steps right and up. One function, $\frac{1}{\sqrt{1-4x}}$, holds all of these numbers at once: the idea of a generating function (next lecture).

A table to remember

Function	Maclaurin series	Valid for
$\frac{1}{1-x}$	$\sum x^k$	$-1 < x < 1$
e^x	$\sum \frac{x^k}{k!}$	all x
$\sin x$	$\sum \frac{(-1)^k x^{2k+1}}{(2k+1)!}$	all x
$\cos x$	$\sum \frac{(-1)^k x^{2k}}{(2k)!}$	all x
$\ln(1+x)$	$\sum_{k\geq 1} \frac{(-1)^{k+1} x^k}{k}$	$-1 < x \leq 1$
$\arctan x$	$\sum \frac{(-1)^k x^{2k+1}}{2k+1}$	$-1 \leq x \leq 1$
$(1+x)^\alpha$	$\sum \binom{\alpha}{k} x^k$	$-1 < x < 1$

Sums start at $k = 0$ unless shown. At $x = -1$, arctan follows from $\arctan(-x) = -\arctan x$.

04

Working with Taylor series

New series from old — and integrals without antiderivatives

By uniqueness (Lecture 5), any power series we build for f *is* its Taylor series.

$$e^{-x^2} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{k!} \qquad \text{(substitute } -x^2 \text{ into } e^u\text{)}$$

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e^{-x^2} has no elementary antiderivative, but term by term:

$$\int_0^1 e^{-x^2}\,\mathrm{d}x = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(2k+1)} = 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \cdots$$

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An alternating series with decreasing terms: five terms give 0.74749, with error at most the next term, $\frac{1}{1320} < 0.0008$. (The true value is $0.746824\ldots$) This integral is at the heart of probability: the Gaussian integral, Unit 21.

Limits

Replace each function by the start of its series:

- $\displaystyle\lim_{x\to 0}\frac{\sin x - x}{x^3} = \lim_{x\to 0}\frac{-\frac{x^3}{6}+\frac{x^5}{120}-\cdots}{x^3} = -\frac{1}{6}$
- $\displaystyle\lim_{x\to 0}\frac{1-\cos x}{x^2} = \lim_{x\to 0}\left(\frac{1}{2}-\frac{x^2}{24}+\cdots\right) = \frac{1}{2}$ (used in Lecture 2)
- $\displaystyle\lim_{x\to 0}\frac{e^x - 1 - x}{x^2} = \frac{1}{2}$

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> **O-notation**
>
> If $f^{(n+1)}$ is bounded near a, then $f(x) = T_n(x) + O\big((x-a)^{n+1}\big)$ as $x \to a$. For example $\sin x = x - \frac{x^3}{6} + O(x^5)$. Often quicker than L'Hôpital's rule, and it shows *how* the limit is approached.

Asymptotics

How fast does $\left(1+\frac{1}{n}\right)^n$ approach e? With $\ln(1+u) = u - \frac{u^2}{2} + O(u^3)$, $n\ln\left(1+\frac{1}{n}\right) = 1 - \frac{1}{2n} + O\left(\frac{1}{n^2}\right)$, and with $e^u = 1 + u + O(u^2)$:

$$\left(1+\frac{1}{n}\right)^n = e \cdot e^{-\frac{1}{2n}+O(1/n^2)} = e\left(1 - \frac{1}{2n} + O\left(\frac{1}{n^2}\right)\right).$$

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n	$\left(1+\frac{1}{n}\right)^n$	$e\left(1-\frac{1}{2n}\right)$
10	2.59374	2.58237
100	2.70481	2.70469
1000	2.716924	2.716923

The same technique gives Stirling's formula for $n!$ (Unit 8).

05

Summary and practice

Summary

- The **Taylor series** of f at a is $\sum \frac{f^{(k)}(a)}{k!}(x-a)^k$. It equals $f(x)$ exactly when the remainder $R_n(x) \to 0$.
- **Taylor's theorem**: $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$, so $|R_n(x)| \le \frac{M|x-a|^{n+1}}{(n+1)!}$ if $|f^{(n+1)}| \le M$.
- Not every smooth function is analytic: e^{-1/x^2}.
- e^x, sin, cos for all x; $\ln(1+x)$ and arctan including $x = 1$: $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$, $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$.
- Binomial series: $(1+x)^\alpha = \sum \binom{\alpha}{k} x^k$ for $|x| < 1$.
- Uses: approximations with error bounds, integrals without antiderivatives, limits, asymptotics.

Practice

1. Find the Maclaurin series and its radius: (a) xe^x (b) $\cos(2x)$ (c) $\sinh x = \frac{e^x - e^{-x}}{2}$ (d) $\ln \frac{1+x}{1-x}$
2. Find the Taylor series of $\ln x$ at $a = 1$. For which x does it converge to $\ln x$?
3. Compute $\cos 0.2$ with an error below 10^{-6}, and justify the bound.
4. Find $\displaystyle\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.
5. Approximate $\int_0^1 \sin(x^2)\,\mathrm{d}x$ with an error below 10^{-3}.
6. Explain why the Taylor series of e^{-1/x^2} at 0 does not represent the function.
7. Show that $\frac{1}{\sqrt{1-4x}} = \sum_{k=0}^{\infty} \binom{2k}{k} x^k$ for $|x| < \frac{1}{4}$.
8. Show that $e^x e^y = e^{x+y}$ by multiplying the series (Cauchy product).

Answers

1. (a) $\sum_{k=0}^{\infty} \frac{x^{k+1}}{k!}$ (b) $\sum_{k=0}^{\infty} \frac{(-1)^k 4^k x^{2k}}{(2k)!}$ (c) $\sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$: all with $R = \infty$. (d) $\ln(1+x) - \ln(1-x) = 2\sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}$, $R = 1$.
2. $\ln x = \ln(1 + (x-1)) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}(x-1)^k}{k}$, for $0 < x \le 2$.
3. $1 - \frac{0.04}{2} + \frac{0.0016}{24} = 0.9800667$. As $T_4 = T_5$ for cos, the error is at most $\frac{0.2^6}{6!} \approx 8.9 \cdot 10^{-8}$.
4. $\arctan x = x - \frac{x^3}{3} + O(x^5)$, so the limit is $\frac{1}{3}$.
5. $\int_0^1 \left(x^2 - \frac{x^6}{6} + \frac{x^{10}}{120} - \cdots\right) \mathrm{d}x = \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} - \cdots$. Then $\frac{1}{3} - \frac{1}{42} \approx 0.3095$, with error below $\frac{1}{1320} < 10^{-3}$ (alternating series).

Answers, continued

6. All derivatives at 0 are 0, so the Taylor series is the zero series. It converges for every x, but to $0 \neq f(x)$ when $x \neq 0$: the remainder $R_n(x) = f(x)$ does not tend to 0.
7. Binomial series with $\alpha = -\frac{1}{2}$ and x replaced by $-4x$ (valid for $|4x| < 1$). Then $\binom{-1/2}{k}(-4)^k = \frac{2^k \cdot 1 \cdot 3 \cdots (2k-1)}{k!} = \frac{(2k)!}{k!\,k!}$.
8. The coefficient of the product $\left(\sum \frac{x^j}{j!}\right)\left(\sum \frac{y^m}{m!}\right)$ collected over $j + m = k$ is $\sum_{j=0}^{k} \frac{x^j y^{k-j}}{j!\,(k-j)!} = \frac{(x+y)^k}{k!}$ by the binomial theorem.

Next time

Lecture 07 — Generating functions (Unit 7)

- A sequence $a_0, a_1, a_2, \ldots$ as the coefficients of one power series
- Solving recurrences, such as the Fibonacci numbers
- Counting with products of series; Project 1

Reading: H. S. Wilf, *generatingfunctionology*, Chapter 1.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.